



Conventional ML Methods

Applied Machine Learning

Classifier Models :: Basic

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Outline

#GeoAI #Mobility #Geography #STDS

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Logistic Regression

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Logistic Regression is the most simplest and linear machine learning classifier among all models.

Although there is a word “**regression**” in the name, this method is used for binary categorization: yes and no, e.g., gender, binary levels, or disease confirmation.

It is a well-known and commonly used method in various fields, especially in medical and sociological studies.

Logistic regression conduct a non-linear function, **sigmoid**, to transform all values as a 0-to-1 scale probability, and we usually adopt 0.5 as a **threshold** for classification.

Logistic Regression

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Suppose we have input features represented as a matrix:

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1m} \\ x_{21} & \cdots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{bmatrix}$$

The dependent variable is Y only a binary value.

$$Y = \begin{cases} 0 & \text{benign} \\ 1 & \text{malignant} \end{cases}$$

Here, we use the multi-linear function as the input variables X .

Logistic Regression

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Here, we use the multi-linear function as the input variables X , like a linear regression (e.g., OLS).

$$z = \sum_{i=1}^m w_i x_i + b = w \cdot X + b$$

where b is bias and $w_i = [w_1, w_2, \dots, w_m]$ is the i th weight or coefficient.

For a classification model, we need to bound the y from 0 to 1, then

$$P(Y = 1|X) = \sigma(z) = \frac{1}{1 + e^{-z}}, \text{ since } 0 \leq P(Y = 1|X) \leq 1$$

Logistic Regression

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Sigmoid Function

The sigmoid function converts the continuous variable data into the probability between 0 and 1.

$\sigma(z)$ tends toward to 1 as $z \rightarrow \infty$

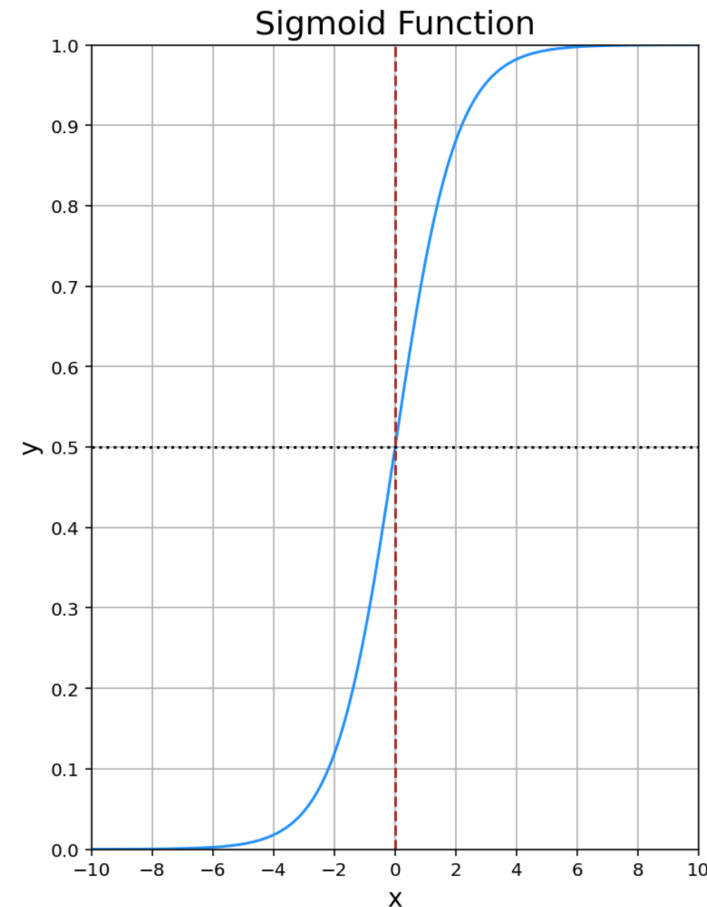
$\sigma(z)$ tends toward to 0 as $z \rightarrow -\infty$

$\sigma(z)$ is bounded between 0 and 1

The probability of each class can be:

$$P(y = 1) = \sigma(z)$$

$$P(y = 0) = 1 - \sigma(z)$$



K-Nearest Neighbors

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K-Nearest Neighbors

What if the relationship between observations cannot be represented well by a simple linear decision boundary?

Here, we introduce another method, called K-Nearest Neighbor Classifier, conducted by identifying the K closest data points to a given input and making predictions based on the majority class or average value of those neighbors.

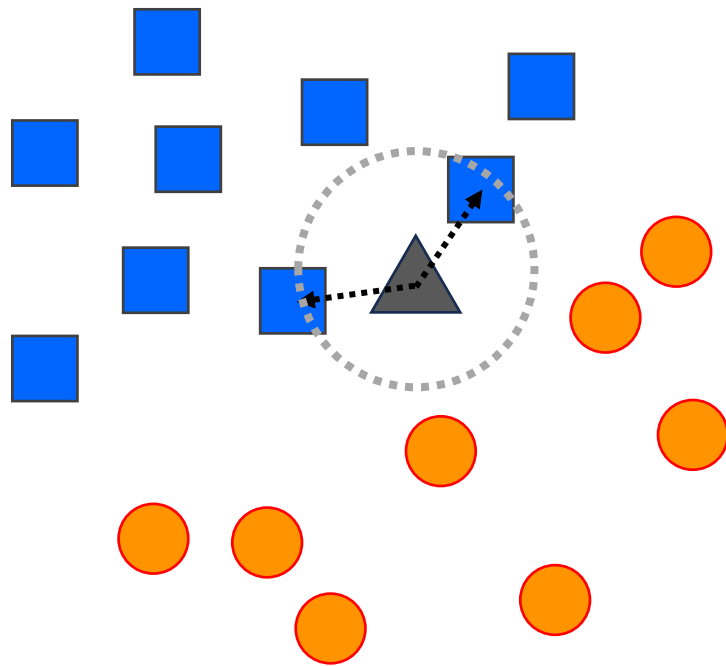
As a result, number of nearest neighbors is the key hyperparameters to significantly affect the model outcome.

→ Similar observations tend to have similar outcomes.

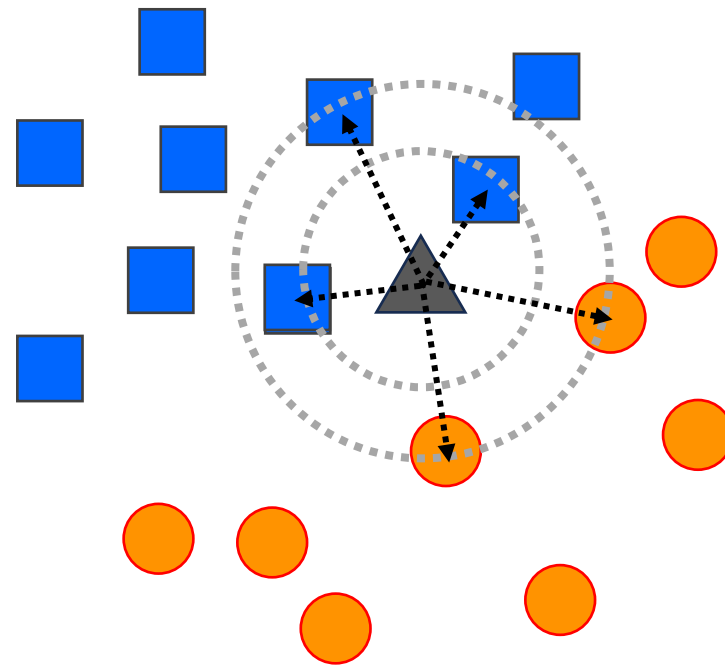
K-Nearest Neighbors

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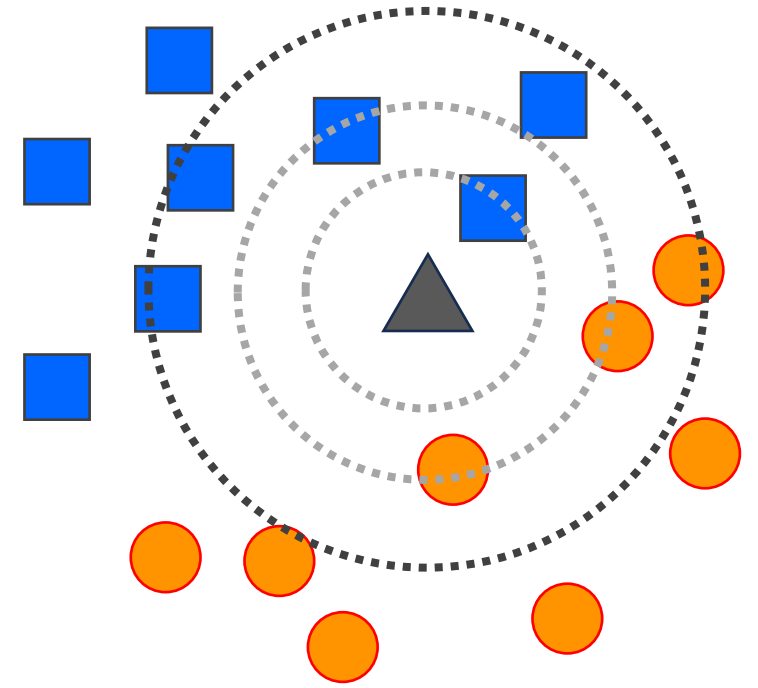
Here, we give three examples ... $K=2, 5, 9$



2 Squares $\rightarrow$ Square



4 Squares + 2 Circles $\rightarrow$ Square



5 Squares + 4 Circles $\rightarrow$ Square

K-Nearest Neighbors

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Hyperparameters

n_neighbors: the number of nearest neighbors

weights: uniform and distance

algorithm: auto, ball_tree, kd_tree, brute

leaf_size: Leaf size passed to BallTree or KDTree

p: Power parameter for the Minkowski metric. When $p = 1$, this is equivalent to using manhattan_distance (L1), and euclidean_distance (L2) for $p = 2$. For arbitrary p , minkowski_distance (L_p) is used.

K-Nearest Neighbors

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Weights

Weight function used in prediction. Possible values:

uniform: uniform weights. All points in each neighborhood are weighted equally.

distance: weight points by the inverse of their distance. in this case, closer neighbors of a query point will have a greater influence than neighbors which are further away.

K-Nearest Neighbors

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Minkowski metric

Metric to use for distance computation. Default is minkowski, which results in the standard Euclidean distance when $p = 2$.

$$d(a, b) = \left(\sum_{i=1}^n |a_i - b_i|^p \right)^{\frac{1}{p}}$$

p=1 Manhattan Distance

p=2 Euclidean Distance

p=∞ Chebyshev Distance

K-Nearest Neighbors

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Algorithms

It determines how the algorithm finds the nearest data points during query time.

Brute Force: Computes the distance from the query point to **every single** sample in the training dataset and sorts them to find the top k .

kd_tree: Recursively splits feature space along axis-aligned **hyperplanes** to build a binary tree.

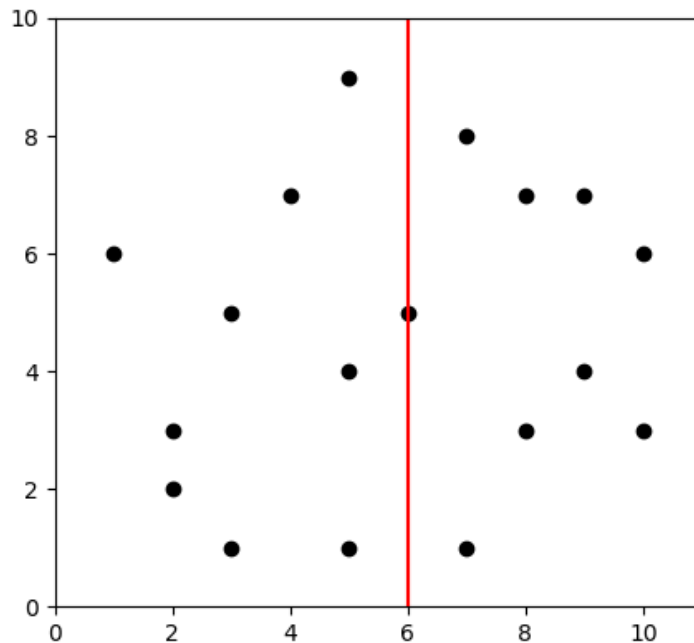
ball_tree: Groups data points into nested D -dimensional **hyperspheres** (balls) defined by a centroid and radius.

K-Nearest Neighbors

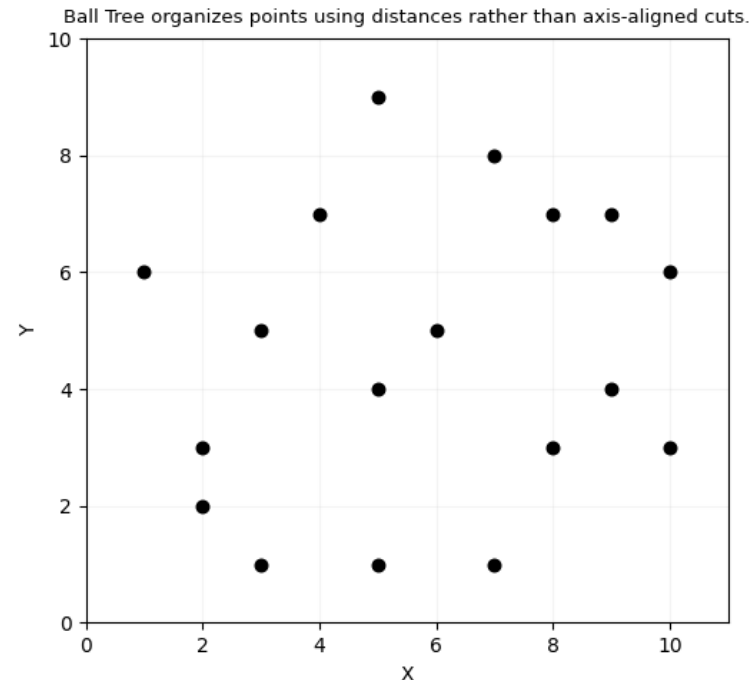
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Algorithms KD Tree

Red = X-axis split
Blue = Y-axis split



1. Original observations



Ball Tree

Tree structure (top → bottom)

Root
n=18

K-Nearest Neighbors

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Summary

KNN does **not learn** a formula or model during training, and **memorizes** the data and decides only during prediction.

We found that as dimensionality increases, the feature space grows exponentially, making the data increasingly sparse.

In high-dimensional spaces, finding enough **nearby** observations requires searching through a surprisingly large portion of the feature space → **Curse of Dimensionality**

Therefore, the differences in the relative distance of nearest and furthest observations could be much smaller.

Naïve Bayes

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Naïve Bayes

Instead of defining similarity geometrically, can we classify observations probabilistically?

Naive Bayes is a machine learning classification algorithm that predicts the category of a data point using probability. It assumes that all features are independent of each other.

Naïve Bayes

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Some Assumptions for Naïve Bayes Classifiers

a) Feature Independence

Assumes every feature contributes to the classification independently, with no interactions or correlations between them.

b) Normally Distributed Continuous Features

Assumes numeric, continuous data follows a Gaussian (normal) bell-curve distribution within each class.

c) Multinomial Discrete Features

Assumes categorical or count data follows a multinomial distribution within each class.

Naïve Bayes

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Some Assumptions for Naïve Bayes Classifiers

d) Equal Feature Weight

Assumes all features carry equal weight and importance when predicting the target class.

e) No Missing Data

Assumes the dataset is complete, with no missing values.

Naïve Bayes

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Bayes' Theorem

It provides a principled way to reverse conditional probabilities.

$$P(y | X) = \frac{P(X | y) \cdot P(y)}{P(X)}$$

$P(y | X)$: Posterior probability, probability of class y given features X

$P(X | y)$: Likelihood, probability of features X given class y

$P(y)$: Prior probability of class y

$P(X)$: Marginal likelihood or evidence

Naïve Bayes

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The **naïve** in Naive Bayes comes from the assumption that all features are independent given the class.

$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) \cdot P(x_2 | y) \cdot \dots \cdot P(x_n | y)$$

Hence, Bayes' theorem becomes

$$P(y | x_1, x_2, \dots, x_n) = \frac{P(y) \cdot \prod_{i=1}^n P(x_i | y)}{P(X)}$$

Since the denominator is a constant for a given input, so

$$P(y | x_1, x_2, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n P(x_i | y)$$

Naïve Bayes

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$$P(y \mid x_1, x_2, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n (x_i \mid y)$$

Thus, we compute the posterior for each class y and choose the class with

$$\hat{y} = \operatorname{argmax}_y P(y) \prod_{i=1}^n (x_i \mid y)$$

And this is Naïve Bayes Classifier.

Naïve Bayes

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Outlook	Yes	No	P(Yes)	P(No)
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rainy	3	2	3/9	2/5
Total	9	5	9/9	5/5

Temp.	Yes	No	P(Yes)	P(No)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
Total	9	5	9/9	5/5

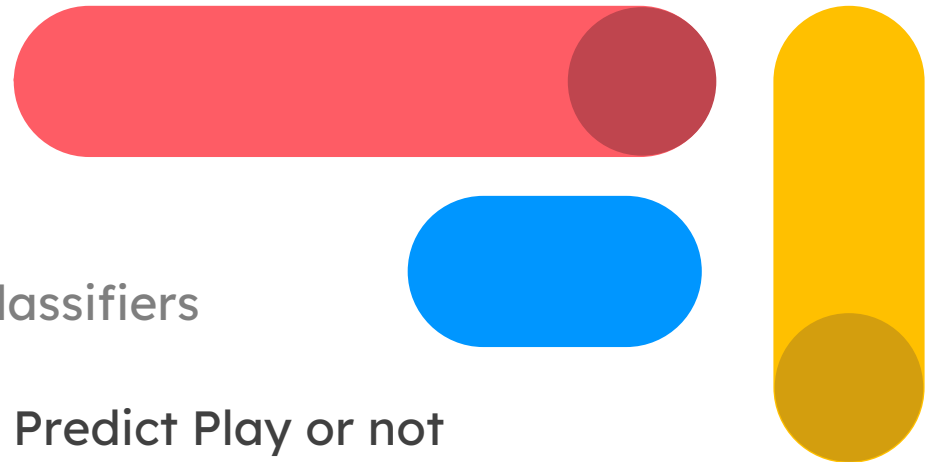
Humidity	Yes	No	P(Yes)	P(No)
Hot	3	4	3/9	4/5
Normal	6	1	6/9	1/5
Total	9	5	9/9	5/5

Wind	Yes	No	P(Yes)	P(No)
False	6	2	6/9	2/5
True	3	3	3/9	3/5
Total	9	5	9/9	5/5

Play	Yes	No	P(Yes)	P(No)
Play	9	5	9/14	5/14

Naïve Bayes

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Outlook	P(Yes)	P(No)
Sunny	2/9	3/5
Overcast	4/9	0/5
Rainy	3/9	2/5

Temp.	P(Yes)	P(No)
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	P(Yes)	P(No)
Hot	3/9	4/5
Normal	6/9	1/5

Wind	P(Yes)	P(No)
False	6/9	2/5
True	3/9	3/5

Play	P(Yes)	P(No)
Play	9/14	5/14

Predict Play or not

Given $X = \{\text{Sunny}, \text{Hot}, \text{Normal}, \text{False}\}$

For Play=Yes

$$P(\text{Yes} | \text{Today}) = \frac{2}{9} \times \frac{2}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14} = 0.0141$$

For Play=No

$$P(\text{No} | \text{Today}) = \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14} = 0.0069$$

Normalize the probability

$$P(\text{Yes} | \text{Today}) = \frac{0.0141}{0.0141 + 0.0069} = 0.673$$

$$P(\text{No} | \text{Today}) = \frac{0.0069}{0.0141 + 0.0069} = 0.327$$

As a result, the prediction will be **Play**

Naïve Bayes

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Depends on posterior distribution, ...

BernoulliNB	Naive Bayes classifier for multivariate Bernoulli models.
CategoricalNB	Naive Bayes classifier for categorical features.
ComplementNB	The Complement Naive Bayes classifier described in Rennie et al. (2003).
GaussianNB	Gaussian Naive Bayes (GaussianNB).
MultinomialNB	Naive Bayes classifier for multinomial models.

Naïve Bayes

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Advantages

- a) **Fast & Simple:** Easy to implement and computationally efficient.
- b) **Scalable & Data-Efficient:** Performs well with high-dimensional features and limited training data.
- c) **Handles Varied Data Types:** Works well with categorical data (and assumes normal distribution for continuous features).

Naïve Bayes

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Disadvantages

Independence Assumption: Assumes features are independent, which rarely holds in real-world data.

Sensitivity to Noise: Performance drops with irrelevant attributes.

Zero-Frequency Problem: Assigns zero probability to unseen categories, hurting generalization.

Decision Tree

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Decision Tree

Can we learn classification rules directly from the data without assuming a specific probability distribution or feature independence?

Decision Trees (DTs) are a non-parametric supervised learning method used for classification

For instance, in the example below, decision trees learn from data to approximate a sine curve with a set of if-then-else decision rules. The deeper the tree, the more complex the decision rules and the fitter the model.

Decision Tree

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How do we determine which feature should be the first and values for splitting?

a) GINI index

$$gini = 1 - \sum_{i=1}^n p_i^2$$

b) Entropy

$$H(S) = - \sum_{i=1}^n p_i \log_2 p_i$$

Urban	Count	Prob
Commerical	3	3/20
Residential	6	6/20
Recreation	5	5/20
Industrial	6	6/20
Total	20	20/20

$$gini = 1 - \left[\left(\frac{3}{20} \right)^2 + \left(\frac{6}{20} \right)^2 + \left(\frac{5}{20} \right)^2 + \left(\frac{6}{20} \right)^2 \right] = 1 - \frac{106}{400} = 0.735$$

$$H(S) = - \left[\frac{3}{20} \log_2 \frac{3}{20} + \frac{6}{20} \log_2 \frac{6}{20} + \frac{5}{20} \log_2 \frac{5}{20} + \frac{6}{20} \log_2 \frac{6}{20} \right] = 1.953$$

Decision Tree

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ID	Age	Exercise	Disease
A	25	6	0
B	30	5	0
C	35	4	0
D	40	3	0
E	45	5	0
F	50	2	1
G	55	4	0
H	60	2	1
I	65	1	1
J	70	3	1

Number of samples (N) is 10, where $N_0 = 6$ and $N_1 = 4$.

Here, we use Gini index as an example

$$Gini_{root} = 1 - [p_0^2 + p_1^2] = 1 - \left[\left(\frac{6}{10} \right)^2 + \left(\frac{4}{10} \right)^2 \right] = 0.48$$

Since $Gini_{root}$ is not 0, we start with the following two features.

Let's start with Age and then Exercise.

Decision Tree

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Feature AGE

Values are (25, 30, 35, 40, 45, 50, 55, 60, 65, 70)

Using the mid-value between two neighboring values could be 27.5, 32.5, 37.5, ..., 67.5

Taking $\text{Age} \leq 47.5$ as an example, we got

ID	Age	Disease
A	25	0
B	30	0
C	35	0
D	40	0
E	45	0

$$\begin{aligned} N_L &= 5 & N_R &= 5 \\ Gini_L &= 1 - (1^2 + 0^2) = 0 & Gini_R &= 1 - (0.8^2 + 0.2^2) = 0.32 \\ Gini_{split} &= \frac{N_L}{N} Gini_L + \frac{N_R}{N} Gini_R = \frac{5}{10} \times 0 + \frac{5}{10} \times 0.32 = 0.16 \\ \text{Impurity reduction is} \\ \Delta Gini &= Gini_{root} - Gini_{split} = 0.48 - 0.16 = 0.32 \end{aligned}$$

ID	Age	Disease
F	50	1
G	55	0
H	60	1
I	65	1
J	70	1

Decision Tree

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Feature Exercise

Taking $Exercise \leq 2.5$ as an example, we got

ID	Age	Disease
F	2	1
H	2	1
I	1	1

$$N_L = 3$$

$$N_R = 7$$

$$Gini_L = 1 - (1^2 + 0^2) = 0 \quad Gini_R = 1 - \left(\frac{1^2}{7} + \frac{6^2}{7} \right) = 0.245$$

$$Gini_{split} = \frac{N_L}{N} Gini_L + \frac{N_R}{N} Gini_R = \frac{3}{10} \times 0 + \frac{7}{10} \times 0.245 = 0.171$$

Impurity reduction is

$$\Delta Gini = Gini_{root} - Gini_{split} = 0.48 - 0.171 = 0.309$$

$$\Delta Gini_{age} > \Delta Gini_{exercise}$$

So, we choose age with $Age \leq 47.5$ as the first split.

ID	Age	Disease
A	6	0
B	5	0
C	4	0
D	3	0
E	5	0
G	4	0
J	3	1

Decision Tree

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Therefore, we can formulize as ...

For each feature: $X_1, X_2, \dots, X_n$, then candidate splits are $s_1, s_2, \dots$

We may compute Gini index for each split

$$I_{split} = \frac{N_L}{N} I_L + \frac{N_R}{N} I_R$$

Then, we obtain the impurity and impurity reduction to decide the split candidate

$$(X_j^*, s^*) = \operatorname{argmin}_{X_j, s} I_{split} \Rightarrow \operatorname{argmax}_{X_j, s} \Delta I$$

Decision Tree

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Hyperparameters

Criterion: The function to measure the quality of a split, i.e., Gini and entropy.

Splitter: The strategy used to choose the split at each node, i.e., best and random.

max_depth: The maximum depth of the tree.

Decision Tree

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Hyperparameters

min_samples_split: The minimum number of samples required to split an internal node.

min_samples_leaf: The minimum number of samples required to be at a leaf node.

min_weight_fraction_leaf: The minimum weighted fraction of the sum total of weights (of all the input samples) required to be at a leaf node.

Decision Tree

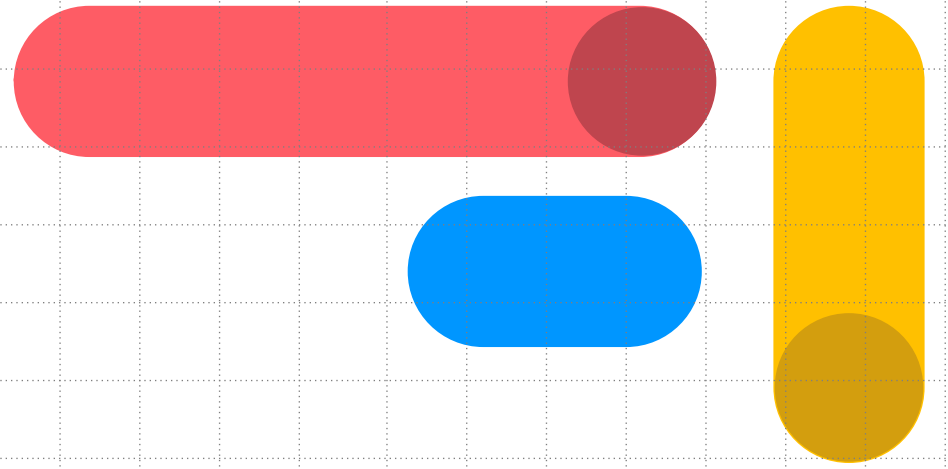
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Disadvantages

Decision Tree has a nature of **Low Bias** and **High Variance**; therefore, if there is a small changes in training dataset, the tree might generate totally different.

Here come the first question: How can we reduce the high variance of a single decision tree?

So, instead of relying on one unstable tree, what if we train many trees and combine their predictions?



The End

Thank you for your attention!

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